

GPT-6.1 SOL

10MAA

End of Year Revision Examination 2026

Student's full name: _____

Form class: _____

Maths teacher's code: _____

Time allowed: 2 hours 25 minutes

Instructions:

- Write in blue/black pen only. Use a pencil for diagrams.
- No twink or correction fluid allowed.
- Calculators are permitted.
- Answer **all questions in this booklet**, showing working in the appropriate space.
- Unless otherwise stated, give non-exact answers to **three significant figures**, angles to **one decimal place**, and money to **two decimal places**.

Topic	**Mark**
Algebraic Expressions & Equations	/31
Statistics & Probability	/27
Trigonometry	/13
Measurement	/16
Coordinate Geometry	/12
Graphs of Quadratic Functions	/23
Geometry	/38
Total	**/160**

Useful Formulae

The Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Volume of a cylinder

$$V = \pi r^2 h$$

Volume of a sphere

$$V = \frac{4}{3}\pi r^3$$

Volume of a cone

$$V = \frac{1}{3}\pi r^2 h$$

The Cosine Rule

$$a^2 = b^2 + c^2 - 2bc\cos A$$

The Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The Area of a Triangle

$$A = \frac{1}{2}ab\sin C$$

The midpoint of a line segment

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Distance

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Gradient of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Equation of a line

$$y - y_1 = m(x - x_1)$$

Perpendicular criteria

$$m_1 m_2 = -1$$

For lines with defined gradients.

Algebraic Expressions & Equations — 31 marks

1. A student writes $(x-4)^2 = x^2 + 16$.

   (a) Expand $(x-4)^2$ correctly.

   Answer: ..... [2]

   (b) Simplify $\dfrac{x^2-16}{x-4}$ for $x \neq 4$.

   Answer: ..... [2]

   (c) The student's original equality is true for exactly one value of x. Find that value.

   x: ..... [2]

2. (a) Simplify $\left(16m^{12}\right)^{\frac{1}{4}}$, where $m > 0$.

   Answer: ..... [2]

   (b) The numbers a and b satisfy

   $$9^a \times 27^b = 3^{22}, \quad a = b + 1.$$

   Find a and b.

   Answer: ..... [3]

Algebraic Expressions & Equations continued

3. A compiler processes files in batches. The table gives the time T, in milliseconds, for a batch containing n files.

n	1	2	3	4
T / milliseconds	7	14	25	40

(a) Find the first differences and show that the second differences are constant.

[2]

(b) Find a quadratic formula for T in terms of n.

T: ..... [3]

(c) Use your formula to predict the time for a batch of 10 files.

Answer: ..... milliseconds [1]

4. A coding club buys 20 stickers for \$78. Python stickers cost \$3 each and Rust stickers cost \$5 each.

(a) Let p be the number of Python stickers and r the number of Rust stickers. Form two simultaneous equations.

[2]

(b) Find how many stickers of each type the club buys.

Answer: ..... [2]

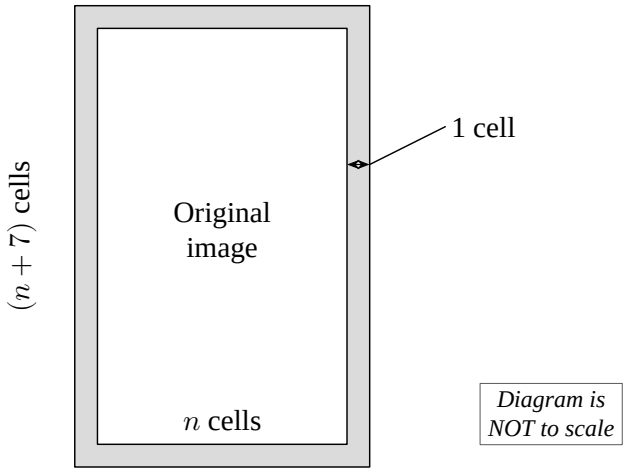
5. The inequality $2(3x - 5) > 5x + k$ is true when $x = 4$, but false when $x = 1$.

k is an integer. Find all possible values of k.

k: ..... [4]

6. A rectangular pixel image is n cells wide and $(n + 7)$ cells high.

A border one cell thick is added around all four sides. The enlarged image, including the border, contains 198 cells.



(a) Form an equation in n.

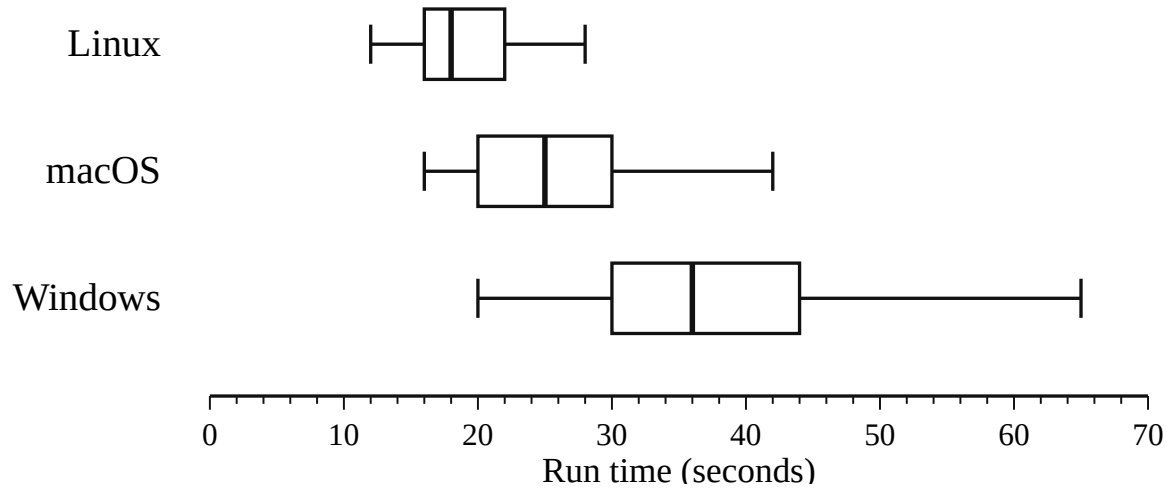
Answer: ..... [2]

(b) Find the dimensions of the original image. Justify your choice of solution.

Answer: ..... cells [4]

Statistics & Probability — 27 marks

1. The boxplots summarise a fictional benchmark. The same set of 20 tasks is run on each operating system.



(a) Find the difference between the median Windows and Linux run times.

Answer: ..... seconds [1]

(b) Which of Linux and macOS has the smaller interquartile range? Show your calculations.

Answer: ..... [2]

(c) Estimate the percentage of runs taking less than 30 seconds for:

macOS: …….. Linux: …….. [2]

(d) Do the boxplots establish that every Linux run was faster than every Windows run? Give a reason.

[1]

2. Six values, written in ascending order, are

$$2, \quad 5, \quad 7, \quad 9, \quad k, \quad 15.$$

Their mean is 8.

(a) Find k.

k: ..... [2]

(b) The final value should have been 150, not 15. State the median before and after this correction, and explain your answer.

[2]

3. Nine cards are labelled with programming languages:

   4 Python cards, 3 Rust cards and 2 C cards.

   Two cards are selected at random without replacement.

   (a) Draw a probability tree showing whether each selected card is Rust or not Rust. Label every branch with its probability.

   [3]

   (b) Calculate the probability that the two cards show the same programming language.

   Answer: ..... [3]

   (c) Given that the cards show the same language, find the probability that both are Rust.

   Answer: ..... [2]

   (d) The experiment is repeated 180 times, with all nine cards returned before each new pair is selected. Find the expected number of pairs showing the same language.

   Answer: ..... pairs [1]

4. The table records the build times for 72 Rust projects.

Build time, t / seconds	Frequency
$0 < t \leq 20$	12
$20 < t \leq 40$	18
$40 < t \leq 80$	24
$80 < t \leq 140$	18

Two students use the table to draw the diagrams below.

Diagram A

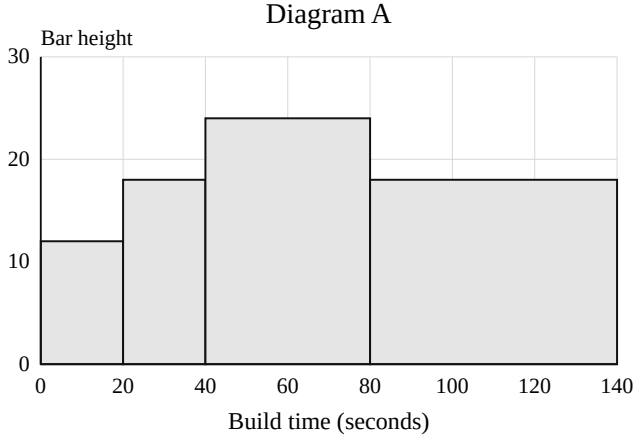
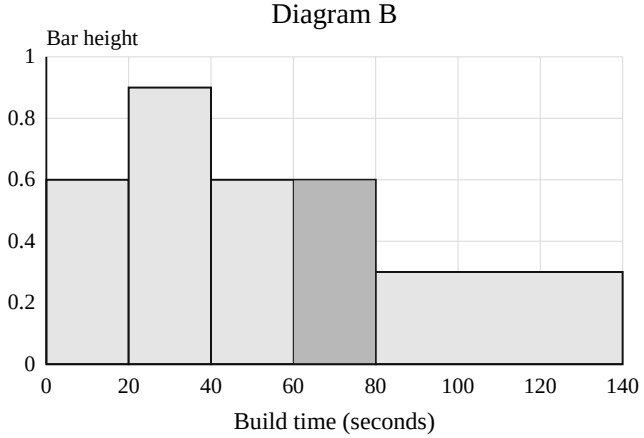


Diagram B



(a) Which diagram is a correct histogram? Justify your answer using bar areas.

[2]

(b) Estimate how many projects are represented by the darker shaded region in Diagram B. State the assumption used for your estimate.

Answer: ..... projects [2]

(c) Without assuming an even spread within classes, find the least and greatest possible numbers of projects with build times greater than 50 seconds.

Answer: ..... [2]

(d) New caching halves every build time. State the new class interval and frequency density for the bar that originally represented $40 < t \leq 80$.

Answer: ..... [2]

Trigonometry — 13 marks

1. A triangle ABC has $AB = 14$ cm, $AC = 10$ cm and $\angle ABC = 35°$.

   There are two possible positions for C, shown as C_1 and C_2.

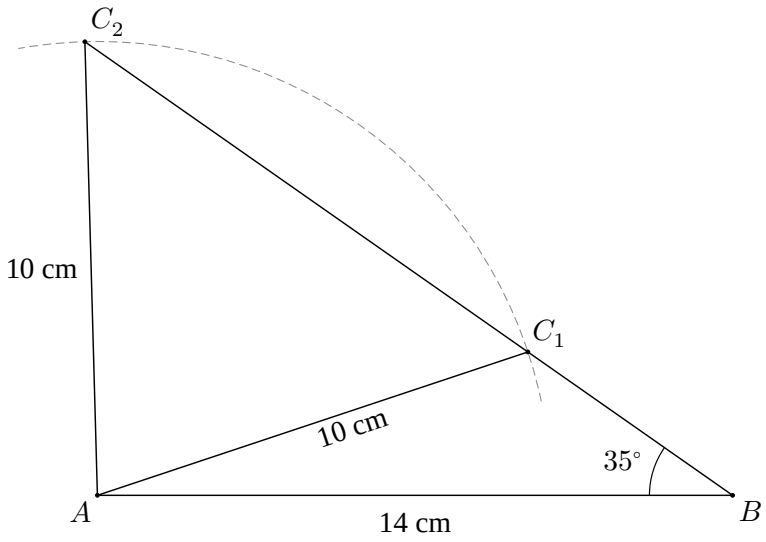


Diagram is NOT to scale

(a) Find both possible values of angle ACB.

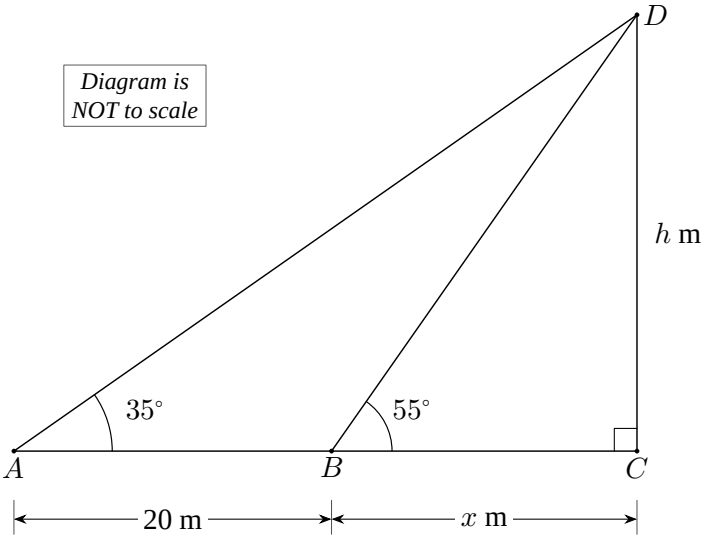
Answer: ..... ° [3]

(b) For the triangle in which angle ACB is obtuse, calculate BC.

BC: ..... cm [2]

2. A, B and C lie on horizontal ground. A mast CD is vertical. The angle of elevation of D is $35°$ from A and $55°$ from B.

$AB = 20$ m, $BC = x$ m and $CD = h$ m.



(a) Express h in terms of x, using the angle at B.

h: ..... [1]

(b) Use the angle at A to form an equation in x only.

Answer: ..... [1]

(c) Calculate the height of the mast.

Answer: ..... m [3]

Trigonometry continued

3. A delivery drone flies 8 km from A to B on a bearing of 040°, then 6 km from B to C on a bearing of 100°.

   Calculate the direct distance from A to C.

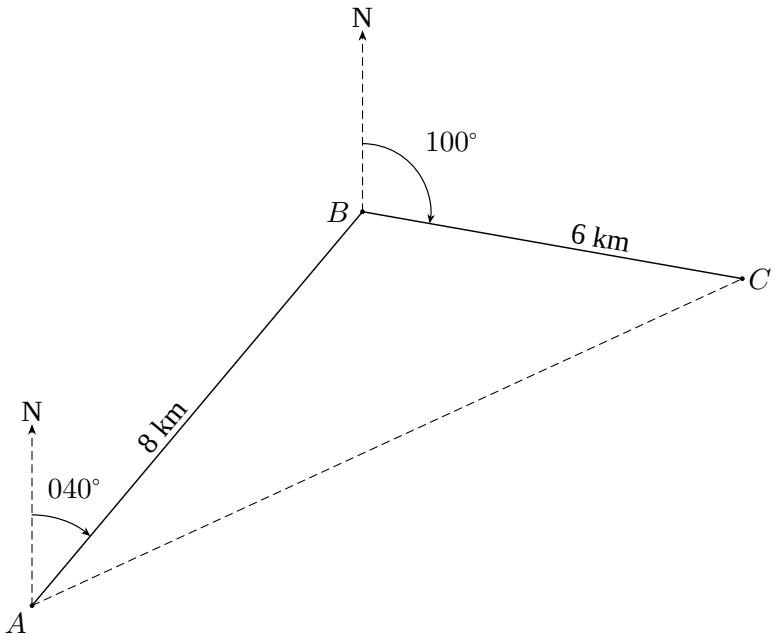


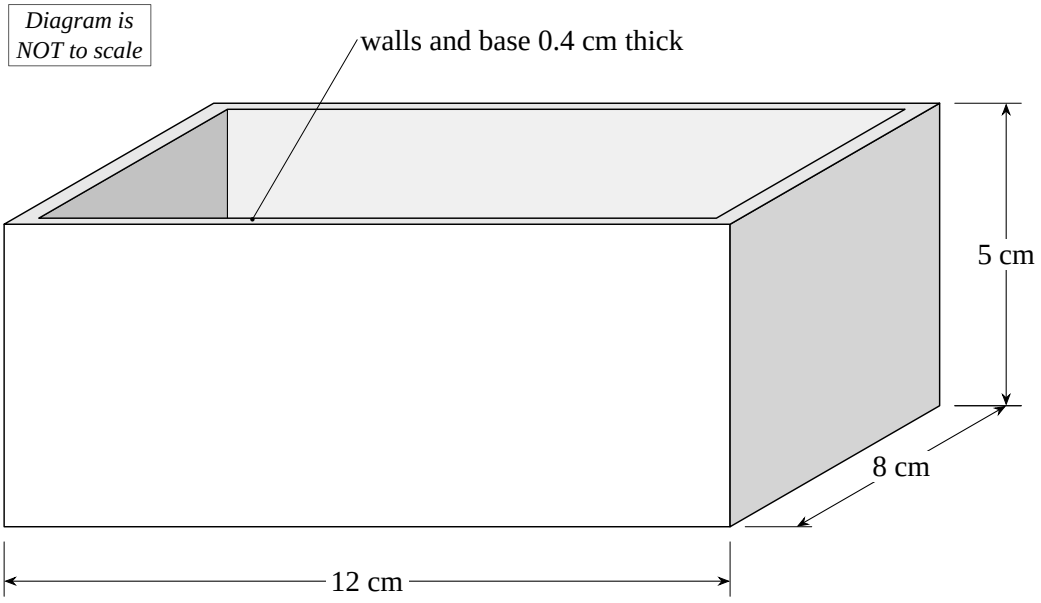
Diagram is NOT to scale

AC: ..... km [3]

Measurement — 16 marks

1. A 3D-printed case is an open-topped rectangular box. Its external dimensions are 12 cm by 8 cm by 5 cm.

   The four walls and the base are each 0.4 cm thick.



   (a) Calculate the internal capacity of the case.

Answer: ..... cm³ [3]

   (b) Calculate the volume of material used to print the case.

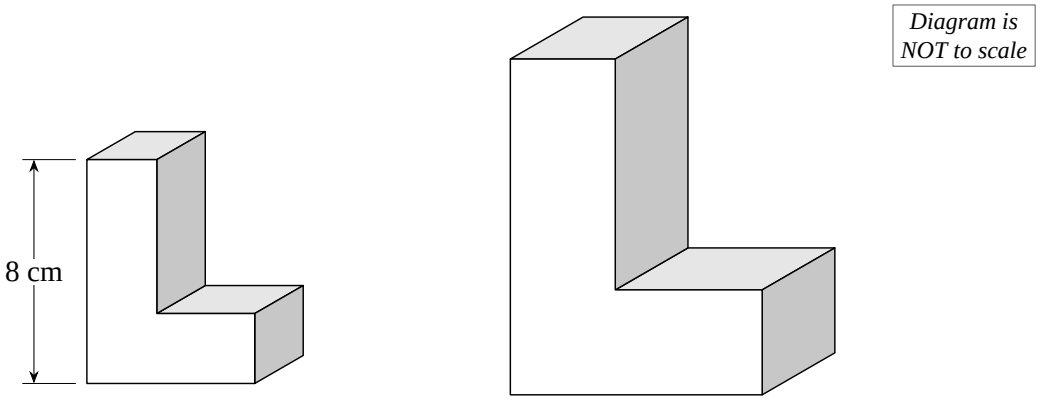
Answer: ..... cm³ [2]

   (c) The printing material has density $1.24\text{g}/\text{cm}^3$. Calculate the mass of the case.

Answer: ..... g [1]

2. Two cooling ducts are geometrically similar. Their volumes are 216cm^3 and 729cm^3. The smaller duct is 8 cm high.

Diagram is NOT to scale



216 cm^3 729 cm^3

(a) Calculate the height of the larger duct.

Answer: ..... cm [2]

(b) An external surface on the smaller duct has area 96cm^2.

Calculate the area of the corresponding surface on the larger duct.

Answer: ..... cm² [2]

3. A cylindrical canister has internal radius 3.0 cm, correct to the nearest 0.1 cm, and internal height 12 cm, correct to the nearest centimetre.

   (a) Write the bounds for the radius r and height h.

   Answer: ..... [2]

   (b) A seller claims that every canister with these measurements can hold at least 350cm^3.

   Use a calculation to decide whether this claim is guaranteed.

   Answer: ..... [2]

   (c) Use the stated values $r = 3.0$ cm and $h = 12$ cm for this part.

   A solid cone has the same radius and height as the canister. The canister is filled with water and the cone is fully inserted, as shown. Water overflows.

   Calculate the volume of water remaining inside the canister.

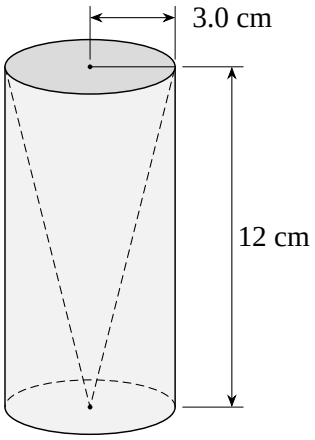
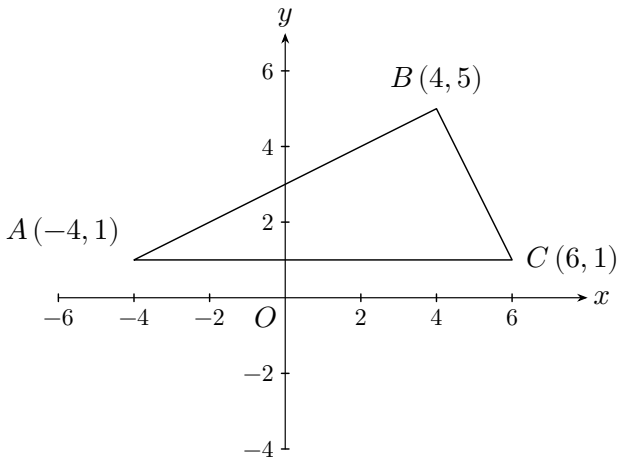


Diagram is NOT to scale

   Answer: ..... cm³ [2]

Coordinate Geometry — 12 marks

1. The diagram shows A $(-4, 1)$, B $(4, 5)$ and C $(6, 1)$.



(a) Show, using gradients, that AB is perpendicular to BC.

[3]

(b) Calculate the area of triangle ABC.

Answer: ..... units² [2]

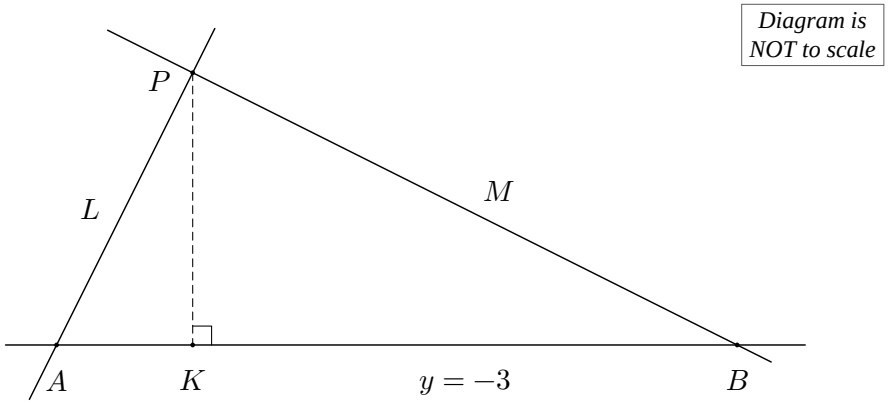
(c) B is reflected in the line AC. Write the coordinates of its image.

Answer: ..... [1]

Coordinate Geometry continued

2. The lines $L : y = 2x + 1$ and $M : y = -\frac{1}{2}x + 6$ intersect at P.

They meet the horizontal line $y = -3$ at A and B respectively. PK is perpendicular to AB.



(a) Find the coordinates of P algebraically.

Answer: ..... [2]

(b) Calculate the length AB.

AB: ..... units [2]

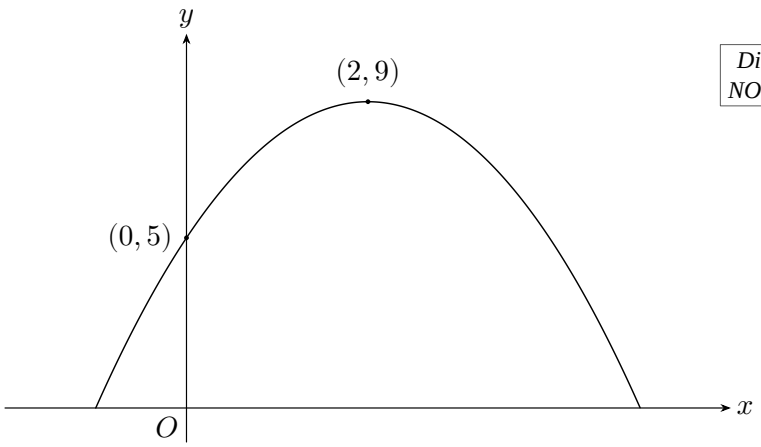
(c) Find the ratio of the area of triangle APK to the area of triangle PKB. Give your answer in its simplest form.

Answer: ..... [2]

Graphs of Quadratic Functions — 23 marks

1. The cross-section of an arch is modelled by a parabola. Its highest point is $(2, 9)$ and it passes through $(0, 5)$.

   y is the height in metres above the floor. x is the horizontal position in metres relative to a reference post. Negative values of x are positions west of the post.



   (a) Find the equation of the parabola in the form $y = a(x - h)^2 + k$.

   Answer: ..... [3]

   (b) Find the horizontal positions where the arch meets the floor.

   Answer: ..... m [3]

   (c) Find the interval of horizontal positions for which the arch is at least 8 m above the floor.

   Answer: ..... [3]

2. An AI program processes n items in a batch. Its memory use, M, in megabytes, is modelled by

$$M = 2n^2 + 6n + 12.$$

n is a positive whole number.

(a) Write $2n^2 + 6n + 12$ in completed-square form.

Answer: ..... [2]

(b) The program has 200 MB of memory available. Find the greatest number of items it can process in one batch. Show how your answer follows from the quadratic model.

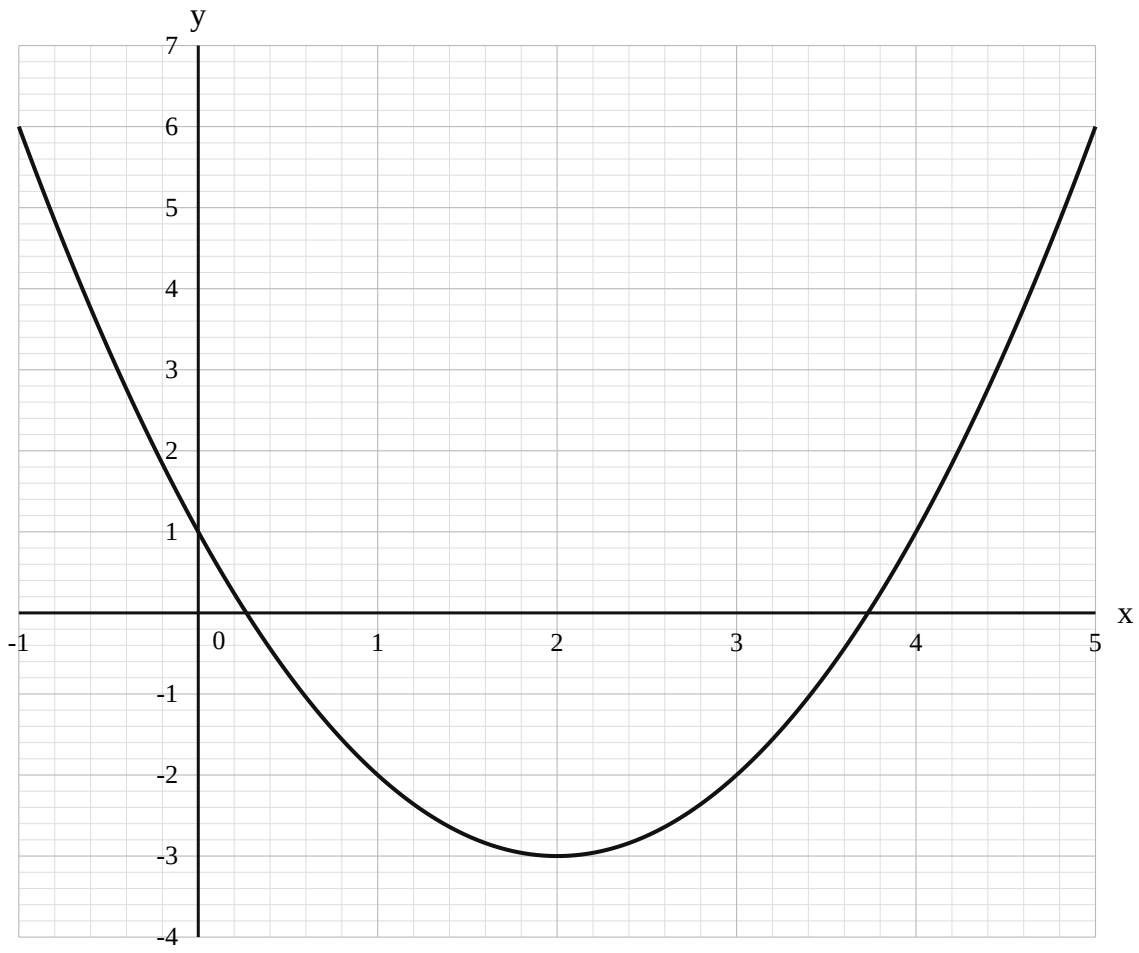
Answer: ..... items [4]

(c) Describe the translation that maps the graph of $y = 2x^2$ onto the graph of $y = 2x^2 + 6x + 12$.

[2]

3. The graph of $y = x^2 - 4x + 1$ is drawn below.

(a) On the same grid, draw the line $y = 1 - x$. [2]



(b) Use the graph to estimate the coordinates of both intersection points.

Answer: ..... [2]

(c) Verify the x-coordinates of the intersections algebraically.

[2]

Geometry — 38 marks

1. TA and TB are tangents to a circle with centre O. A and B are the points of contact. Angle AOB is 110°.

   C lies on the minor arc AB.

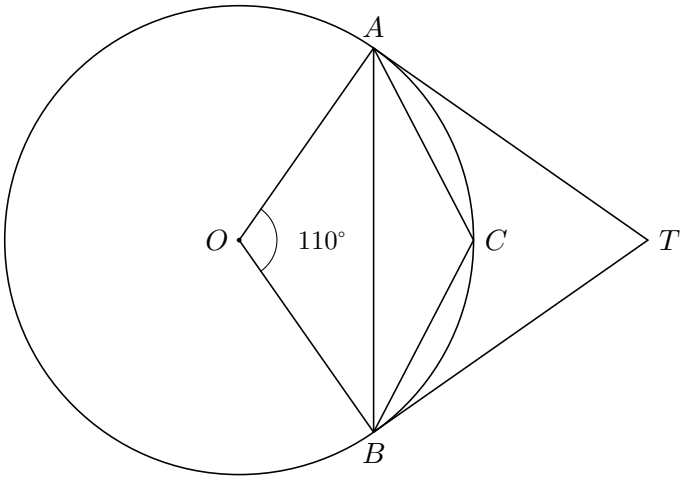


Diagram is NOT to scale

(a) Calculate angle OAB. Give a geometric reason.

[2]

(b) Calculate angle ATB. Give geometric reasons for your steps.

[3]

(c) Calculate angle ACB. Give a geometric reason.

[2]

2. $\overrightarrow{OA} = \boldsymbol{p}$ and $\overrightarrow{OB} = \boldsymbol{q}$.

C lies on AB, with $AC : CB = 2 : 1$.

D is defined by $\overrightarrow{OD} = 2\boldsymbol{q} - \boldsymbol{p}$.

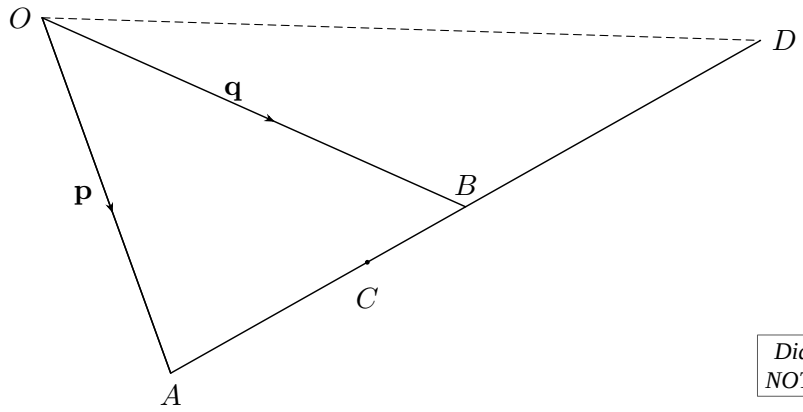


Diagram is NOT to scale

(a) Find $\overrightarrow{OC}$ in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$.

Answer: ..... [2]

(b) Prove, using vectors, that A, C and D lie on one straight line.

[2]

(c) Find the ratio $AC : CD$ in its simplest form.

Answer: ..... [2]

3. In triangle OAB, $\overrightarrow{OA} = \boldsymbol{p}$ and $\overrightarrow{OB} = \boldsymbol{q}$.

P lies on OA with $OP : PA = 2 : 1$. Q lies on OB with $OQ : QB = 2 : 1$.

R is the midpoint of AQ, and S is the midpoint of BP.

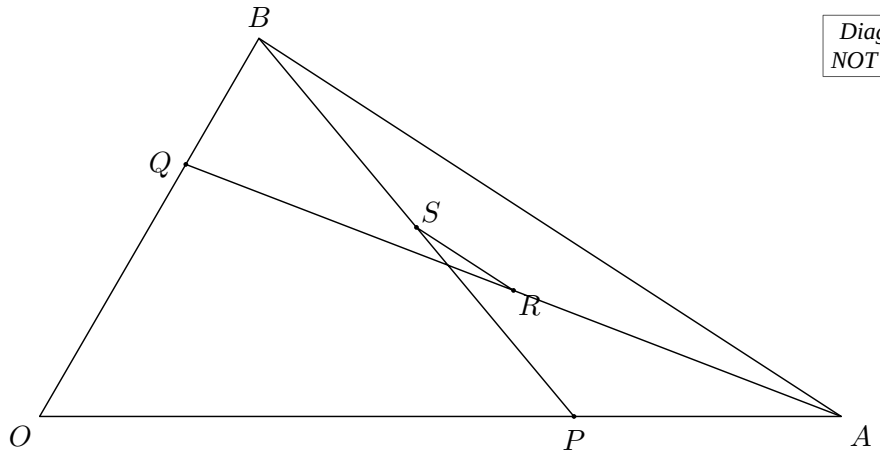


Diagram is NOT to scale

(a) Find $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$.

Answer: ..... [2]

(b) Find $\overrightarrow{OR}$ and $\overrightarrow{OS}$ in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$.

Answer: ..... [2]

(c) Use vectors to prove that RS is parallel to AB, and find $RS : AB$.

[2]

Geometry continued

4. ABCD is a parallelogram, with $\overrightarrow{AB} = \boldsymbol{a}$ and $\overrightarrow{AD} = \boldsymbol{b}$.

P, Q, R and S are the midpoints of AB, BC, CD and DA respectively.

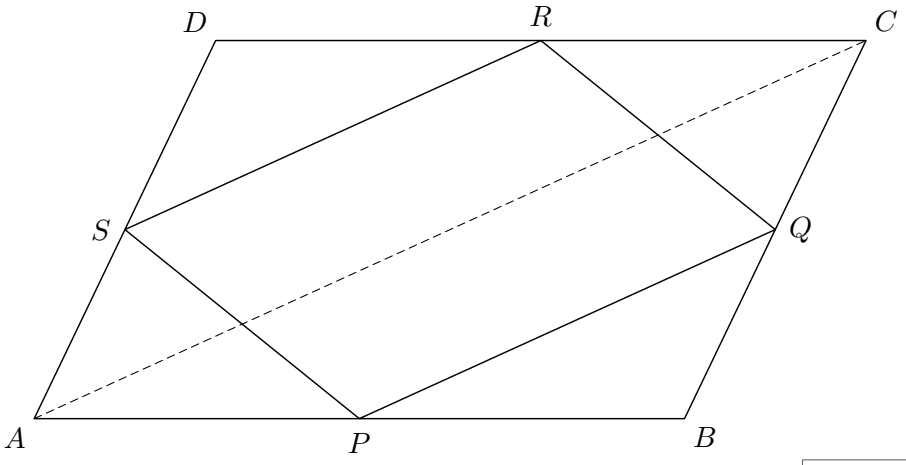


Diagram is NOT to scale

(a) Find $\overrightarrow{PQ}$ and $\overrightarrow{SR}$ in terms of $\boldsymbol{a}$ and $\boldsymbol{b}$.

Answer: ..... [2]

(b) Use your answers to prove that PQRS is a parallelogram.

[2]

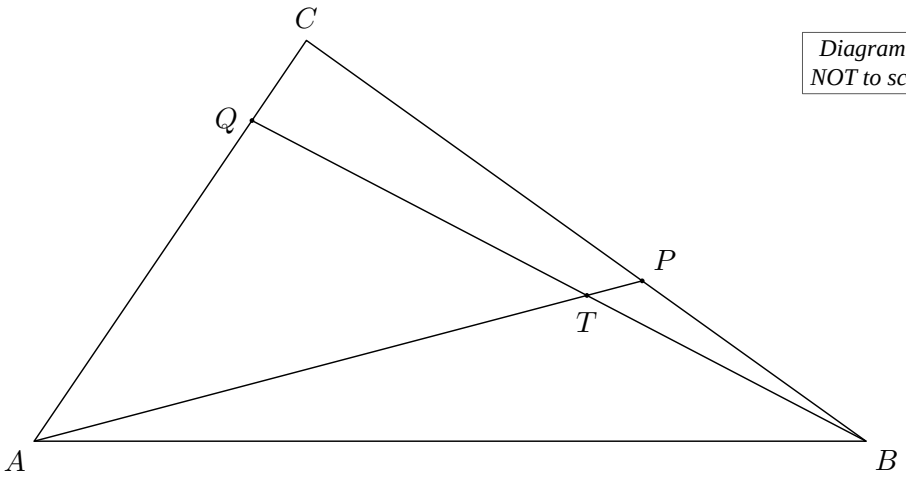
(c) Show, using position vectors from A, that both diagonals of PQRS have the same midpoint as AC.

[2]

5. In triangle ABC, $\overrightarrow{AB} = \boldsymbol{a}$ and $\overrightarrow{AC} = \boldsymbol{b}$.

P lies on BC with $BP : PC = 2 : 3$. Q lies on AC with $AQ : QC = 4 : 1$.

AP and BQ meet at T.



(a) Find $\overrightarrow{AP}$ and $\overrightarrow{BQ}$ in terms of $\boldsymbol{a}$ and $\boldsymbol{b}$.

Answer: ..... [2]

(b) Let $\overrightarrow{AT} = k\overrightarrow{AP}$ and $\overrightarrow{BT} = m\overrightarrow{BQ}$.

Express $\overrightarrow{AT}$ in two ways. By comparing coefficients of $\boldsymbol{a}$ and $\boldsymbol{b}$, form two equations in k and m.

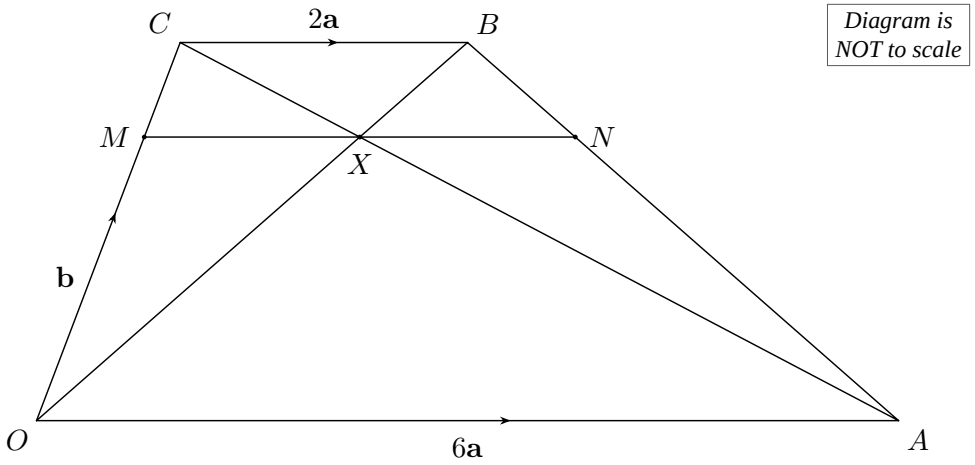
[2]

(c) Find the ratio $AT : TP$.

Answer: ..... [2]

6. OABC is a trapezium, with $\overrightarrow{OA} = 6\boldsymbol{a}$, $\overrightarrow{CB} = 2\boldsymbol{a}$ and $\overrightarrow{OC} = \boldsymbol{b}$.

The diagonals OB and AC meet at X. The line through X parallel to OA meets OC at M and AB at N.



(a) Find $\overrightarrow{OB}$ and $\overrightarrow{AC}$ in terms of $\boldsymbol{a}$ and $\boldsymbol{b}$.

Answer: ..... [2]

(b) Let $\overrightarrow{OX} = s\overrightarrow{OB}$ and $\overrightarrow{AX} = t\overrightarrow{AC}$.

By expressing $\overrightarrow{OX}$ in two ways, show that $s = t = \frac{3}{4}$.

[3]

(c) Find $\overrightarrow{MX}$ and $\overrightarrow{XN}$. Hence prove that X is the midpoint of MN.

[2]

End of Assessment